

OneGemmy

Entity Relationship Diagram & Multi-Tenancy Architecture

FastAPI + SQLAlchemy + SQLite | Feature-Grouped API v1

48 Permissions | 7 Roles | 5 Demo Users

Gemmy Connect Ltd

Built with care in Rwanda

Tenant Hierarchy

Platform	[super_admin]	Manages all companies, users. Global access. No company_id filter.
Company	[company_admin]	Tenant boundary. Owns shops, roles, employees. Scoped to company_id.
Shop	[user]	Operational unit within company. Owns products, transactions. Scoped to shop_id.
User	[—]	Belongs to a company + shop + role. Role determines permission set.
Role	[—]	Per-company or system role. Links to 48 granular permissions via M2M.
Permission	[—]	Atomic access unit: module.action (e.g. stock.view, sales.create).

Data Isolation Strategy

- All business tables (Product, Deal, Lead, Contact, Transaction, Project, Employee) are scoped via company_id.
- Users see only their company's data. Super admins bypass all filters.
- Roles are per-company (company_id = NULL for system roles).
- Shops are per-company. A user can be assigned to one or more shops.
- The get_current_user() dependency loads user + role + permissions on every request.
- The require_permission() dependency checks the loaded permissions list. Returns 403 if missing.

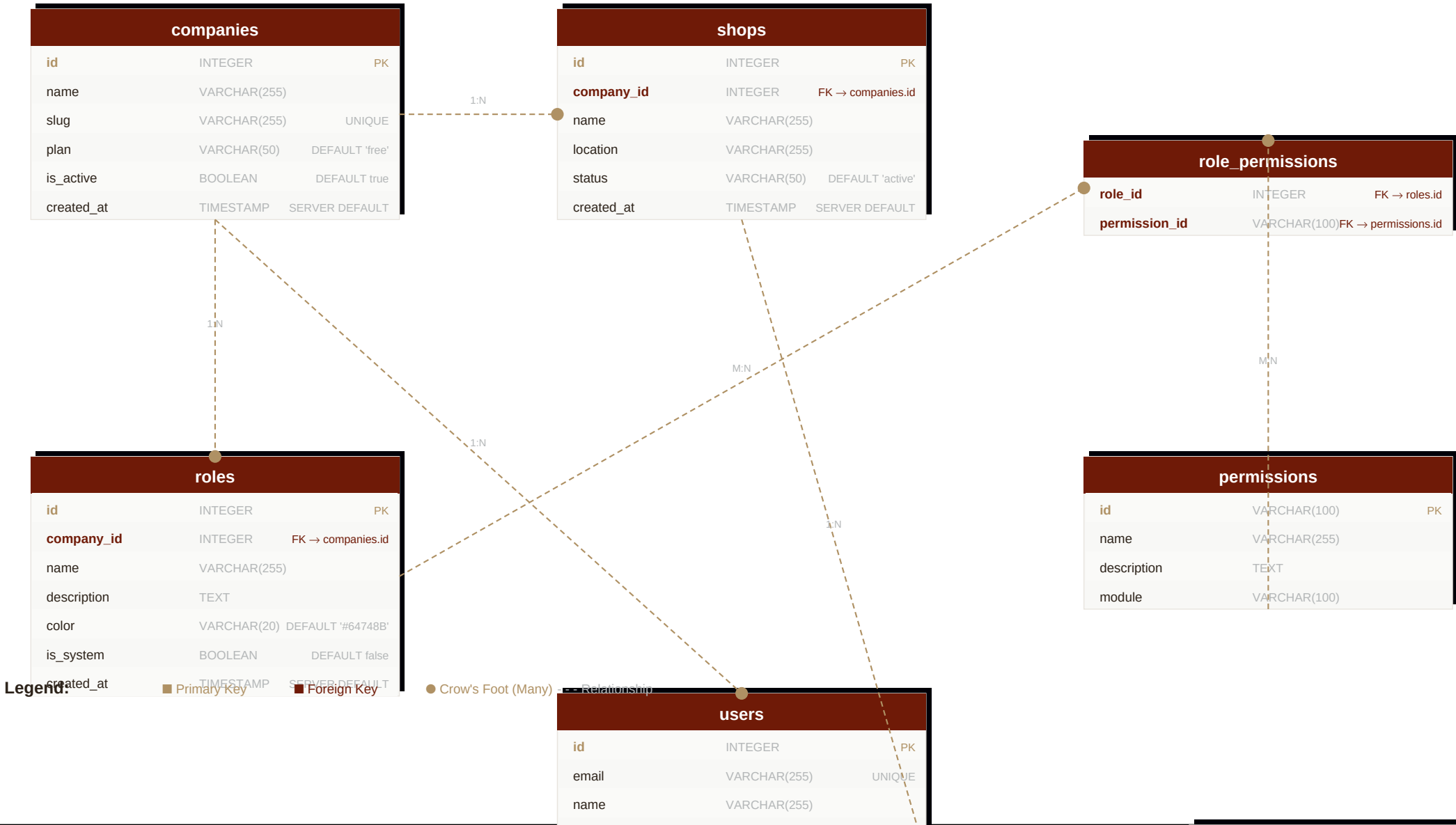
Request Flow (RBAC)

1. Client sends JWT token in Authorization header.
2. get_current_user() decodes token → loads User from DB.
3. require_permission("stock.view") checks: user.role.permissions contains "stock.view"?
4. If yes → route handler runs. If no → 403 Forbidden.
5. Super admins bypass all permission checks (platform_role == "super_admin").
6. Company admins get Owner role permissions (all 48 perms).

Seed Data (Demo)

Roles: Owner (48), Admin (40), Manager (30), Sales Rep (10), Inventory Mgr (12), Accountant (14), Employee (8)
Users: super@onegemmy.com (super_admin), admin@onegemmy.com (Owner), manager@onegemmy.com (Manager),

Entity Relationship Diagram



RBAC Permission Matrix

Role	Perms	Modules	Description
Owner	48	All	Full access to everything. Company creator.
Admin	40	All except user mgmt	Manages everything except inviting users.
Manager	30	Sales, Stock, Finance, Projects	Day-to-day operations across modules.
Sales Rep	10	CRM, Sales (view/create/edit)	Manages leads, deals, contacts.
Inventory Mgr	12	Stock (full), Products	Manages products, stock levels.
Accountant	14	Finance (full), Reports	Manages transactions, invoices, reports.
Employee	8	Projects (view), Contacts (view)	Limited read access. No sensitive data.

Permission Modules (48 total)

dashboard	view
sales	view, create, edit, delete, leads.manage, quotes.manage
stock	view, create, edit, delete
finance	view, create, edit, delete
projects	view, create, edit, delete
crm	view, contacts.manage, interactions.log
hr	view, employees.manage
reports	view
settings	view, users.manage, roles.manage